Therefore: $P(E|F) = \frac{P(E \cap F)}{P(F)} = \frac{\frac{1}{8}}{\frac{1}{2}} = \frac{1}{4}$

1.4 Proofs

1.5 Visualizations

1.6 Practice Problems

Family with Two Children: Assume each child is equally likely to be a boy or girl. If a family has two children, what is the conditional probability that both are girls given that: (i) the youngest is a girl (ii) at least one is a girl

1.7 Worked Solutions

Basic Calculation: If $P(A) = \frac{7}{13}$, $P(B) = \frac{9}{13}$, and $P(A \cap B) = \frac{4}{13}$, then:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{4}{13}}{\frac{9}{13}} = \frac{4}{9}$$

Urn Problem: An urn contains 10 black and 5 white balls. Drawing two balls without replacement: - $P(\text{first black}) = \frac{10}{15}$ - $P(\text{second black} \mid \text{first black}) = \frac{9}{14}$ - $P(\text{both black}) = \frac{10}{15} \times \frac{9}{14} = \frac{3}{7}$

1.8 Analogies

2. Expert Explanation

Conditional probability is one of the most important concepts in probability theory. It measures the probability of an event occurring given that another event has already occurred.

Key Intuition: When we know that event F has happened, we essentially "shrink" our sample space to only include outcomes where F occurs. Within this reduced sample space, we then calculate the probability of event E.

The Formula Breakdown: - $P(E \cap F)$ = probability that both E and F occur - $P(F)$ = probability that F occurs (our "given" condition) - The division gives us the proportion of F-outcomes that also satisfy E

Why P(F) ≠ 0?: We need this condition because we can't condition on an impossible event. If F never happens, the concept of "given F has occurred" is meaningless.

3. Extensions

Real-World Application: Consider medical testing: - E = "person has disease" - F = "test is positive" - $P(E|F)$ = probability person has disease given positive test

This is different from $P(F|E)$ = probability of positive test given disease!

Connection to Independence: Two events E and F are independent if and only if $P(E|F) = P(E)$. This means knowing F occurred doesn't change the probability of E.

4. Next Steps

To deepen your understanding: 1. Practice calculating conditional probabilities with different examples 2. Explore Bayes' theorem, which uses conditional probability 3. Study the multiplication rule: $P(E \cap F) = P(E|F) \times P(F)$ 4. Learn about the law of total probability 5. Apply conditional probability to real-world scenarios like medical diagnosis, quality control, or weather prediction